

different cells altogether. This way you can see an overall view of the chip while zooming in one or more than one piece of the chip, and you can precisely align large components. Additionally, it enables you to easily copy information from one window to the other.

Interactive router

A key feature and an aid to manual routing in this layout tool is its interactive router, Irouter, which provides an interface to the internal maze router. You can perform routing at one connection at a time, specifying the starting and the end point prior to each connection.

What's more, you can decide the order in which signals should be routed and how multi-point nets should be decomposed into point-to-area connections. But the interactive router is not a fully automated router, so it will not consider the interactions between nets and thus net decomposition is not automatic.

The special Magic 'hint' layers will allow you to control the nature of the routes. Typically, you can determine the overall path of a connection and leave the 'design-rule check' and 'de-touring' around or over minor obstacles to the router tool itself.

Design-rule check with error paints

Magic incorporates a continuous design-rule check feature, which enables you to have an up-to-date picture of design rule errors (if any) in your layout. Therefore it is not necessary to have a complete check over the whole design unless design rules are changed.

New features in the latest version

1. Improved routing through an interactive maze router, which allows you to specify hits to control the routing
2. Accurate path-length extractions are now possible, which is very useful while dealing with high-speed circuits
3. An interface to IRSIM simulator

With the interactive feature of continuous design-rule check, you can easily and immediately figure out the error location. While editing any layout with Magic, the system will automatically check the design rules, as mentioned above. Whether you paint or erase, or move a cell or change any array structure, Magic will recheck the area changed by you in order to check for the violation of any of the layout rules. If rules are violated, little white dots will be displayed in the vicinity of the violation. These white dots or the error paint will stay till you fix the problem.

Circuit extraction in hierarchical designs

Magic uses cell hierarchies to represent a layout. Each cell contains three things: paint, label and sub cell. While some say that a hierarchical structure is not necessary by reasoning that layout can also be represented as flat, a hierarchical structure greatly improves efficiency of the design tools as well as the speed and ease of designing with them.

There is a problem though. In case of cell overlaps, this situation becomes complicated. As each cell uses a separate plane, so information must be combined from these separate planes in case of any overlap. Therefore circuits cannot be extracted hierarchically as it can cause transistors to split

between cells or form (appear?) broken in cell-overlap situations.

Magic, however, allows the cells to overlap as long as the transistor structure is not changed. This means, you can have extracted circuit to be represented hierarchically. The extracted circuit will contain the circuit of the cell, circuits of its sub cells and connections between them.

Though Magic can deal with cell overlapping, extensive use of cell overlaps is not recommended. Extended overlapping will cause the tool to run very slowly and hard to make any modifications.

Stretch and compact cells

Plowing is a special feature that allows you to rearrange the layout without changing the electrical circuit which it represents. The plow operation preserves the design rules and connectivity to maintain the electrical structure of the circuit.

As the plow moves, every edge in its path is pushed ahead of it (not the layers), preserving design rules, connectivity, transistors and contact sizes. This makes everything ahead of the plow to get compacted down to a minimum spacing as per the design rules, and the material which crosses its original position gets stretched behind the plow. To get a better idea, you can try it out using the installer in the DVD accompanying this magazine.

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